
WEEK 01 – DAY 01 HOMEWORK

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1. Study of AI Model Providers and Their Recent Models

Artificial Intelligence has evolved rapidly over the past few years, with several organizations releasing increasingly capable Large Language Models (LLMs). These models differ in terms of reasoning ability, coding performance, multimodal support, context length, pricing, and deployment options. Selecting an appropriate model depends on the intended application, such as software development, enterprise automation, research, education, or customer support.

The table below compares ten prominent AI providers and one of their recently released flagship models.

Provider	Model	Release	Context Window	Primary Strength
OpenAI	GPT-5.6 Sol	Jul 2026	1.05M	Advanced reasoning and coding
Anthropic	Claude Opus 5	Jul 2026	1M	Enterprise AI and long-context reasoning
Google DeepMind	Gemini 3.6 Flash	Jul 2026	1M	Fast multimodal processing
xAI	Grok 4.5	Jul 2026	500K	Programming and technical reasoning
Alibaba	Qwen3.8-Max	Aug 2026	1M	Long-context multilingual AI
Moonshot AI	Kimi K3	Jul 2026	1M	Research and AI agents
DeepSeek	DeepSeek-V4-Flash	Apr 2026	1M	Cost-effective inference

Provider	Model	Release	Context Window	Primary Strength
Cohere	Command A+	May 2026	128K	Enterprise deployments
Mistral AI	Mistral Large 3	Dec 2025	256K	Open-weight multimodal AI
ByteDance	Seed-OSS-36B	Aug 2026	512K	Self-hosted AI applications

Overview of Leading AI Providers

OpenAI

OpenAI develops advanced language models capable of reasoning, coding, document analysis, and conversational AI. GPT-5.6 Sol extends previous generations by supporting larger context windows and improved multi-step reasoning. It is widely used in education, software development, research, and enterprise automation.

Anthropic

Anthropic focuses on building reliable and safety-oriented language models. Claude Opus 5 is optimized for enterprise environments where long documents, careful reasoning, and trustworthy responses are essential.

Google DeepMind

Google DeepMind integrates language understanding with multimodal capabilities. Gemini models process text, images, audio, and video within a unified architecture, making them suitable for search, productivity tools, and AI assistants.

xAI

xAI develops models designed to assist programmers, researchers, and engineers. Grok emphasizes technical reasoning, software engineering workflows, and integration with developer tools.

Alibaba – Qwen

Qwen models support multilingual communication, coding assistance, and research applications. Their long-context capability allows efficient processing of lengthy documents and conversations.

Moonshot AI

Moonshot AI specializes in long-context reasoning and autonomous AI agents. Kimi models are particularly effective when multiple reasoning steps or large knowledge bases are involved.

DeepSeek

DeepSeek offers powerful models at relatively low inference costs. These models are increasingly used in Retrieval-Augmented Generation (RAG), semantic search, and enterprise AI solutions where cost efficiency is important.

Cohere

Cohere develops enterprise-focused language models with strong multilingual capabilities. Organizations frequently use these models for customer support, document search, and private deployments.

Mistral AI

Mistral AI provides open-weight models that can be deployed on-premises or in private cloud environments. Their models balance performance with lower computational costs.

ByteDance

ByteDance develops open-weight models suitable for research and self-hosted deployments. These models enable organizations to build customized AI applications without relying entirely on cloud-based APIs.

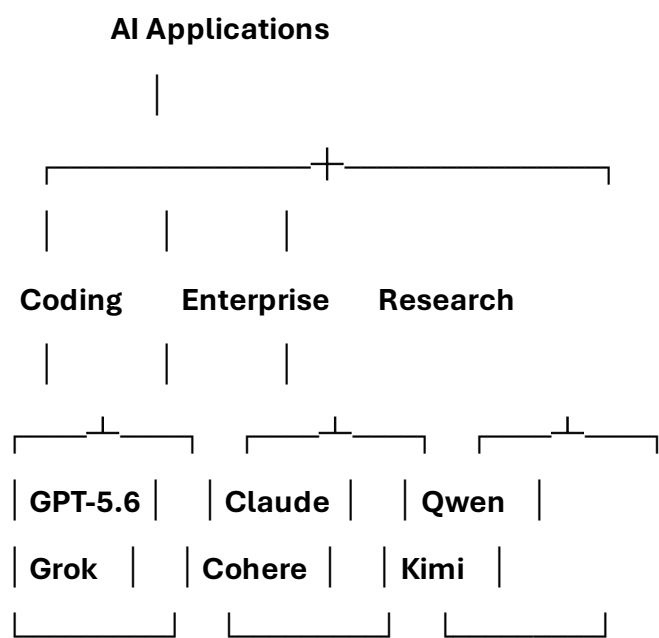
Comparison of AI Models

Each provider focuses on different priorities:

- **OpenAI emphasizes reasoning and coding performance.**
- **Anthropic prioritizes reliability and enterprise use.**
- **Google excels in multimodal AI.**
- **Qwen performs well in multilingual tasks.**
- **DeepSeek provides excellent value for cost-sensitive deployments.**
- **Mistral offers strong open-weight alternatives.**
- **Cohere focuses on secure enterprise applications.**

Organizations select models based on context length, pricing, deployment requirements, and application domain. Some prioritize accuracy, while others value lower operational costs or the ability to self-host models.

Figure 1. AI Model Ecosystem (to be included in the final PDF)



2. Understanding Transformer Models

Reference

DataCamp – How Transformers Work

2.1 Introduction

Transformer models have become the foundation of modern Artificial Intelligence, particularly in Natural Language Processing (NLP). Introduced in the research paper "**Attention Is All You Need**" (2017), Transformers significantly improved how computers process sequential data such as text, speech, and programming code.

Earlier neural networks, especially Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, processed words one after another. Although effective for short sequences, they struggled with long documents because information from earlier words gradually weakened as the sequence became longer.

Transformers addressed this limitation by processing every token simultaneously using a mechanism known as **Self-Attention**. Instead of reading text sequentially, the model

evaluates relationships among all words in parallel, enabling better contextual understanding while greatly increasing training speed.

Today, nearly every advanced Large Language Model—including GPT, Claude, Gemini, Qwen, Llama, and DeepSeek—is built upon the Transformer architecture.

2.2 Why Transformers Were Developed

The primary objective of the Transformer architecture was to improve machine translation.

Consider the English sentence:

"The student completed the assignment before the deadline."

When translating into another language, the model must understand that **completed** relates to **assignment**, while **deadline** provides time-related context.

Traditional RNN-based systems often lost this context in lengthy sentences.

Transformers solve this by examining every word simultaneously, allowing the model to capture both nearby and distant relationships within the sentence.

This parallel processing improves:

- Translation accuracy
 - Text summarization
 - Question answering
 - Code generation
 - Chatbots
 - Information retrieval
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2.3 Basic Transformer Architecture

A Transformer consists of two primary components.

Encoder

The encoder receives the input sentence.

Its responsibilities include:

- Tokenizing text

- Converting tokens into embeddings
- Learning contextual relationships
- Producing contextual representations

Decoder

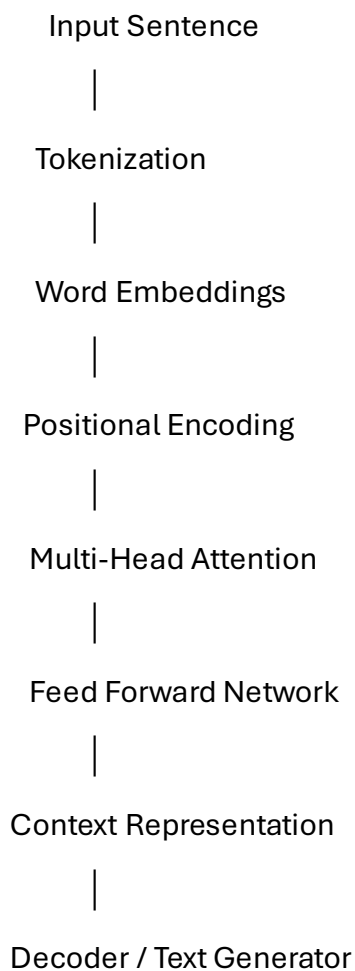
The decoder receives information from the encoder.

Its tasks include:

- Predicting the next token
- Generating fluent output
- Producing complete sentences

Many modern language models, such as GPT, primarily use the decoder, while BERT relies mainly on the encoder.

Figure 2.1 Transformer Workflow



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Final Response

2.4 Self-Attention Mechanism

Self-Attention is the most important innovation introduced by Transformers.

Rather than examining only neighboring words, the model determines which words are most relevant when interpreting each token.

For example:

"The animal didn't cross the road because **it** was tired."

The model learns that **it** refers to **the animal**, not **the road**.

Self-Attention enables:

- Better contextual understanding
 - Long-distance dependency learning
 - Improved reasoning
 - Parallel computation
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2.5 Multi-Head Attention

Instead of using only one attention calculation, Transformers compute several attention patterns simultaneously.

Each attention head focuses on a different aspect of the sentence.

For example:

Head 1:

- Grammar

Head 2:

- Subject-object relationship

Head 3:

- Semantic meaning

Head 4:

- Long-distance dependencies

These attention heads are combined to create richer contextual representations.

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2.6 Popular Transformer Models

BERT

BERT introduced bidirectional language understanding by considering words from both left and right contexts simultaneously. It is commonly used for classification, search, and question-answering applications.

GPT

GPT uses a decoder-only Transformer architecture that predicts one token at a time. It is highly effective for conversational AI, programming assistance, summarization, and creative writing.

Claude

Claude focuses on safe, reliable, and helpful responses. It is widely adopted for enterprise AI, document analysis, and customer support.

Gemini

Google's Gemini combines text, images, audio, and video within a unified Transformer architecture, making it suitable for multimodal AI systems.

2.7 Modern Transformer Variants

Recent research has introduced several improvements.

Vision Transformers (ViT)

Instead of processing words, Vision Transformers divide images into patches.

Each patch behaves similarly to a text token, allowing the model to recognize objects and classify images.

Applications include:

- Medical imaging

- Autonomous vehicles
 - Satellite image analysis
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Multimodal Transformers

Modern AI systems combine multiple data types.

Examples:

- Text
- Images
- Audio
- Video

This allows a single model to answer questions about images, summarize videos, or analyze mixed-content documents.

Flash Attention

Flash Attention reduces memory usage while increasing computational speed.

Benefits include:

- Faster inference
 - Lower GPU memory
 - Longer context windows
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Rotary Position Embeddings (RoPE)

RoPE improves positional encoding, helping Transformers process much longer documents without losing contextual information.

Mixture of Experts (MoE)

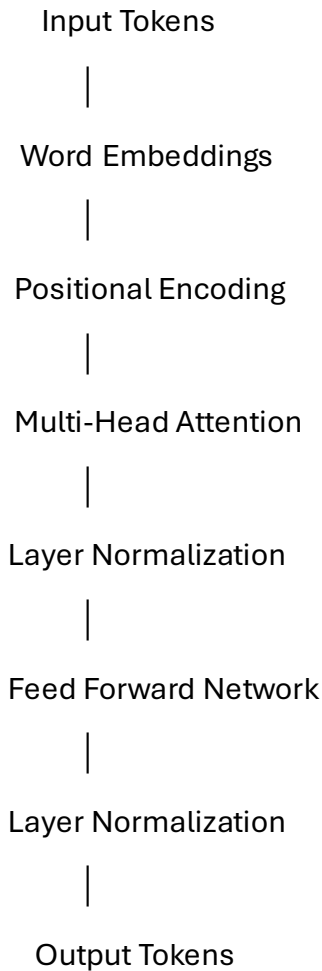
Instead of activating every neural network component, MoE activates only the most relevant expert networks for each token.

Advantages:

- Faster computation

- Lower cost
 - Larger model capacity
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Figure 2.2 Modern Transformer Architecture



Summary

Transformer models have transformed Artificial Intelligence by enabling efficient parallel processing and superior contextual understanding. Their flexibility allows them to power applications ranging from conversational AI to computer vision and multimodal reasoning. Continuous innovations such as Flash Attention, RoPE, and Mixture of Experts have made modern Transformers faster, more scalable, and capable of handling increasingly complex real-world tasks.

3. Vector Embeddings and Retrieval-Augmented Generation (RAG)

Reference

AWS – Retrieval-Augmented Generation (RAG)

3.1 Introduction to Vector Embeddings

Large Language Models cannot directly understand human language in the form of words or sentences. Before processing text, every sentence must be converted into a numerical representation called an **embedding**.

An embedding is a vector containing hundreds or thousands of floating-point numbers that mathematically represent the meaning of the text.

Unlike keyword-based approaches, embeddings capture **semantic similarity**.

For example:

"I bought a new car."

"I purchased a new automobile."

Although both sentences use different words, their meanings are almost identical.

Embedding models therefore generate vectors that are positioned close together within vector space.

This property makes embeddings the foundation of modern AI search systems.

3.2 How Embeddings Are Generated

The embedding generation process consists of several stages.

1. Raw text is received.
2. The sentence is tokenized into smaller pieces.
3. Each token is converted into numerical representations.
4. Transformer layers analyze relationships between all tokens.
5. The model produces one high-dimensional vector representing the complete sentence.

Typical embedding dimensions include:

- 384

- 768
- 1024
- 1536
- 3072

Higher-dimensional vectors generally preserve richer semantic information.

Figure 3.1 Embedding Generation Process

User Query

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Sentence

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Tokenizer

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Transformer Encoder

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1024-Dimensional Vector

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Vector Database

3.3 Semantic Search

Traditional keyword search compares exact words.

For example,

Searching:

Reduce cloud cost

may fail to retrieve

Optimizing AWS Spending

because none of the keywords match directly.

Semantic search solves this limitation.

Instead of comparing words, it compares embeddings.

If two vectors are close in vector space, the documents are considered semantically similar.

Advantages include:

- Better search accuracy
 - Better recommendation systems
 - Faster information retrieval
 - Improved document discovery
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3.4 Retrieval-Augmented Generation (RAG)

One major limitation of Large Language Models is that they only know information learned during training.

Retrieval-Augmented Generation (RAG) improves accuracy by retrieving external knowledge before generating a response.

The workflow is:

1. Documents are split into chunks.
2. Each chunk is converted into embeddings.
3. Embeddings are stored inside a vector database.
4. User questions are embedded.
5. Similar document chunks are retrieved.
6. Retrieved information is supplied to the LLM.
7. The model produces an answer using both retrieved knowledge and its own reasoning.

This greatly reduces hallucinations and allows AI systems to answer questions using organization-specific knowledge.

Figure 3.2 Retrieval-Augmented Generation Pipeline

Documents

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Chunking

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Embeddings

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Vector Database

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Similarity Search

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Retrieved Context

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Large Language Model

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Final Answer

3.5 Popular Vector Databases

Several vector databases are widely used in modern AI applications.

Pinecone

A cloud-native vector database optimized for semantic search and Retrieval-Augmented Generation.

Milvus

An open-source vector database capable of managing billions of vectors efficiently.

MongoDB Atlas Vector Search

Extends MongoDB by adding vector similarity search capabilities alongside traditional document storage.

Redis

Supports fast in-memory vector similarity search for real-time applications.

Elastic

Provides hybrid search combining keyword search and vector search.

Real-World Applications

Vector embeddings are widely used in:

- Chatbots
 - Recommendation systems
 - Image retrieval
 - Medical document search
 - Enterprise knowledge management
 - AI Agents
 - Customer support automation
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4. Feedforward Neural Networks (FNN)

Reference

GeeksforGeeks – Feedforward Neural Network

4.1 Introduction

A Feedforward Neural Network (FNN) is one of the simplest architectures in deep learning. Information flows in a single direction—from the input layer through one or more hidden layers to the output layer. Unlike recurrent models, there are no feedback loops, making FNNs suitable for tasks where the entire input is available at once.

4.2 Network Architecture

An FNN consists of three main layers:

Input Layer

Receives the input features (for example, age, income, or house size).

Hidden Layers

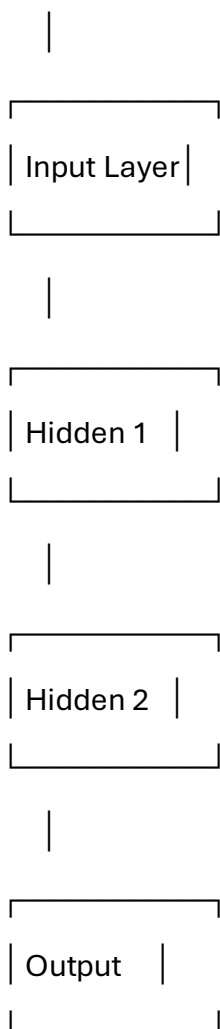
Each hidden neuron computes a weighted sum of its inputs, adds a bias term, and applies an activation function. Multiple hidden layers allow the network to learn increasingly complex patterns.

Output Layer

Produces the final prediction. Depending on the task, this could be a class label (classification) or a continuous value (regression).

Figure 4.1 Feedforward Neural Network

Input Features



4.3 Activation Functions

Activation functions introduce non-linearity, allowing the network to model complex relationships.

- **Sigmoid:** Produces outputs between 0 and 1, commonly used for binary classification.
 - **Tanh:** Produces outputs between -1 and 1, often leading to faster convergence than Sigmoid.
 - **ReLU:** Outputs the input directly if positive; otherwise returns zero. ReLU is the most widely used activation function because of its simplicity and effectiveness.
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4.4 Training Process

Training an FNN follows these steps:

1. **Forward Propagation:** Input data moves through the network to generate predictions.
2. **Loss Calculation:** The prediction is compared with the actual value using a loss function such as Mean Squared Error (MSE) or Cross-Entropy Loss.
3. **Backpropagation:** Errors are propagated backward to compute gradients.
4. **Gradient Descent:** Weights are updated iteratively to reduce the prediction error.

During inference, only the forward propagation step is required because the learned weights remain fixed.

Practical Applications

Feedforward Neural Networks are used in:

- Image classification
 - Speech recognition
 - Credit scoring
 - Fraud detection
 - Medical diagnosis
 - Recommendation systems
 - Sales forecasting
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Summary

Vector embeddings enable AI systems to compare information based on meaning rather than exact wording, making semantic search and RAG possible. Feedforward Neural Networks provide the foundation for many supervised learning tasks by learning patterns through layered computation. Together with Transformer models, these techniques form the core technologies behind modern AI systems.

5. Conclusion

Artificial Intelligence has advanced significantly due to the development of modern deep learning architectures and large language models. Organizations such as OpenAI, Anthropic, Google DeepMind, Alibaba, xAI, Cohere, Mistral AI, and others have introduced increasingly capable models for coding, reasoning, multilingual communication, and multimodal understanding. Each provider focuses on different strengths, allowing users to select models based on their application requirements.

Transformer models have become the backbone of modern Natural Language Processing because they process information using self-attention rather than sequential computation. This enables them to understand relationships between words more effectively and handle very large contexts. Recent improvements, including Vision Transformers, Flash Attention, Rotary Position Embeddings, and Mixture of Experts, continue to improve efficiency and scalability.

Vector Embeddings have transformed information retrieval by representing text as numerical vectors that preserve semantic meaning. Together with Vector Databases and Retrieval-Augmented Generation (RAG), embeddings enable AI systems to retrieve relevant information from external knowledge bases before generating responses. This approach improves factual accuracy while reducing hallucinations.

Feedforward Neural Networks remain one of the fundamental neural network architectures in machine learning. Although newer architectures such as Transformers dominate many NLP applications, FNNs continue to play an important role in classification, regression, computer vision, recommendation systems, and predictive analytics.

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